

Q1. $\phi(x) = \frac{x^2 + 1}{2x - 3}$.

$\phi : X \rightarrow Y$, for some subsets $X, Y \subseteq \mathbb{R}$

Choose X, Y so that we have all four combinations of injectivity, surjectivity and their negations shown.

- Can you identify the values $y \in \mathbb{R}$ that do not occur as values of ϕ i.e. $\forall x \in \mathbb{R} \quad \phi(x) \neq y$.

Let's try to exactly describe relevant aspects of ϕ . Formula for ϕ undefined at $x = \frac{3}{2}$, as $2(\frac{3}{2}) - 3 = 0$ there.

From the plot it appears that $\phi(x)$ avoids zero. Let's describe this gap precisely. An element $y \in \mathbb{R}$ occurs as a value of ϕ if there exists $x \in \mathbb{R}$ such that

$$y = \frac{x^2 + 1}{2x - 3}$$

